Ranking, Calibration and Cost in Competing-Risks Survival Models: A Common-Protocol Benchmark on Mortgage Data

Amirreza Sadraei, Ali Emre Nebiler, Filip Becic
Department of Computer and Systems Sciences, Stockholm University

***Abstract*—Lenders must state how likely a loan is to default by a given date, not merely which loans are riskiest, and a loan can end by default, prepayment or maturity. Deep competing-risks models are usually reported on ranking alone, and usually on splits that let them train on loans originated alongside the loans they are tested on. We benchmark cause-specific Cox, Fine–Gray, a random survival forest and DeepHit on 550,000 Freddie Mac mortgages, measuring discrimination, calibration and cost together under a tuning budget fixed in advance. Under rolling-origin evaluation no model leads reliably: over four time-ordered folds per cohort, every advantage lies within 1.3 standard deviations of its own spread across folds, and a single split had made one of them look significant. Holding the test loans and the training-set size fixed and moving only the training period locates the advantage that remains. Training on contemporaneous loans is worth up to 0.13 AUC, is worth more to DeepHit than to the Cox model in three of four cells, and is the difference between DeepHit leading the forest in every cell and trailing it, a change significant in all four. The calibration penalty, by contrast, does not depend on the protocol: DeepHit's probabilities sit further from observed frequencies in both regimes, and a rescaling estimated on data a lender actually has does not repair it.**

I. Introduction

Credit scoring has long been posed as classification at a fixed horizon: will this borrower default within a year? Provisioning rules ask something harder. A lender must estimate losses over the remaining life of a loan, which requires a probability of default at every future date, and loans end in more than one way. A mortgage may default, be repaid early, or run to maturity, and once one of these happens the others cannot. Under such competing risks the quantity needed is the cumulative incidence function of default rather than one minus a survival curve [1]. The distinction is not academic: treating early repayment as missing information biases the estimated default probability upward [2], and a model can order borrowers correctly while attaching the wrong numbers to them.

Deep models that predict a cumulative incidence per cause exist and have been applied to mortgages [3], [4], but they are selected and reported on ranking. Our three literature reviews — of competing risks in credit scoring, of calibration measures, and of evaluation protocol and cost — converged independently on that omission: these models' probabilities are seldom checked, and the protocols used to check anything seldom resemble deployment. This paper asks whether the advantage survives a protocol that measures ranking, calibration and cost together, on 550,000 Freddie Mac mortgages across two economic regimes.

II. Background and Related Work

Two recent reviews bracket this problem without meeting in the middle. Shang et al. [5] survey three decades of survival analysis in credit risk but do not examine how the resulting predictions are checked. Wiegrebe et al. [6] review more than fifty deep learning methods for time-to-event data and observe that most handle only a single risk with censoring; they consider no credit application. Neither asks whether credit models produce faithful probabilities.

On the credit side, the field's benchmark compares eight survival models on ten portfolios and finds that seven of them censor early repayment; on the same loans that choice moves the censoring rate from 20–85% to 94.56–98.16% [7]. Later work adopts competing risks explicitly, with cause-specific Cox models [8], random survival forests [2] and boosting [9], while macroeconomic covariates have long been used in the single-risk setting [10]. Blumenstock et al. [4] benchmarked DeepHit and a random survival forest against cause-specific Cox and Fine–Gray on 600,000 US mortgages with default and prepayment as competing risks, reporting the highest time-dependent concordance index for DeepHit in every experiment. Their design already splits temporally, with random subsampling inside it into folds of ten thousand instances — a size forced by the forest's compute — and one of their experiments trains

before the crisis and tests in 2009. Time ordering is therefore not what distinguishes the present study. What distinguishes it is the quantity measured: their evaluation is concordance alone, and the words calibration and Brier do not appear in the paper. Our model set and event pair are theirs; we add calibration and cost, train at full scale rather than on subsamples, and measure what the choice of split protocol is itself worth.

That omission is not peculiar to one paper, and it is not always visible as an omission. Saavedra-García et al. [9] apply competing-risks boosting to IFRS 9 lifetime default and do report three measures, but describe the dynamic AUC as "providing a measure of calibration over time" and read the integrated Brier score as indicating "effective calibration". The first is a discrimination measure; the second is a proper score that confounds the two. A recent paper in the target application can therefore believe it has measured calibration while reporting neither a calibration curve nor a calibration error. On the evaluation side, DeepHit's loss combines a likelihood with a ranking term whose weight is tuned on validation discrimination [3]; Haider et al. [11] define D-calibration for single events, and Sonabend et al. [12] show that a concordance index computed on a predicted distribution depends on an unstated reduction to a single risk. Most recently, Alberge et al. [13] show that existing calibration measures are unsuited to competing risks, introduce two proper measures, and give recalibration methods that improve probabilities while preserving discrimination. These measures have not been applied to credit data, and no study evaluates ranking, calibration and cost for these models together.

Two qualifications matter for how that gap should be read. First, the deployed baseline is not a survival model: reviewing bank disclosures, Bank and Eder [14] find lifetime probabilities of default under IFRS 9 built mainly from transformations of regulatory estimates and rating-based matrix models, with discrete-time multinomial logit the usual device where competing risks are modelled at all. Competing risks have therefore reached production, in a model family absent from the benchmarking literature. Second, calibration is not neglected in practice: supervisory guidance defines it as matching a long-run average default rate, and current IFRS 9 term-structure work still censors competing risks for tractability: Botha and Verster [15] report that 47% of the mortgage spells in their study end in a competing risk, 96% of them early settlements, and treat those spells as right-censored. The gap is thus a mismatch between the quantity the benchmarking literature reports and the quantity deployment requires, not an absence of concern for probabilities. Wycinka finds the same on consumer loans: discrimination separated the specifications far less than their agreement with the empirical cumulative incidence did [16].

III. Research Problem and Question

Our three reviews reached the same literature from different directions. A.S. surveyed competing-risks models in credit scoring and found that early repayment is usually censored rather than modelled as a second outcome. A.E.N. surveyed calibration measures for survival predictions and found that credit studies report discrimination almost to the exclusion of anything else. F.B. surveyed evaluation protocol and cost and found that random train/test splits dominate and that fitting cost is seldom reported. These are three symptoms of one condition: deep competing-risks models are chosen and reported on a ranking statistic, under a protocol unlike the way a lender meets a new loan, and their probabilities go unchecked.

The gap follows directly. This research evaluates how much of the reported advantage of deep competing-risks models is a property of the way they are measured rather than of the models themselves. Our research question is: *to what extent is the reported advantage of deep competing-risks models over classical ones an artefact of the protocol used to evaluate them?* It divides into four sub-questions. Does any model lead reliably once the time-ordered split is not a single draw (§V-A)? How much of a reported lead comes from training on loans originated alongside the test loans (§V-B)? What calibration penalty accompanies it, and how much of that penalty can recalibration repair (§V-C)? And what does the deep model cost (§V-D)?

IV. Protocol

The strategy is experimental in Denscombe's sense: an empirical investigation under controlled conditions producing numerical outcomes, with hypotheses fixed before measurement and controls held over every factor not under test. It is closer to an evaluation design than to a treatment experiment. The data are secondary records of loans rather than of people, so no ethical approval was required; the licence permits academic use but not redistribution, which is why the repository excludes the data and rebuilds the tables from archives the reader downloads.

Four delimitations were fixed before any model was run: US agency mortgages only; covariates observed at origination rather than time-varying macroeconomic series; horizons up to 60 months rather than a full mortgage term; and the lifetime PD term structure alone,

one input to expected credit loss, leaving loss given default, exposure at default, discounting and staging out of scope.

A. Data

Our data are 50,000-loan samples per origination year from the Freddie Mac Single-Family Loan-Level Dataset: 300,000 loans originated 2013–2018 and 250,000 originated 2005–2009, with performance observed to March 2026. The post-crisis cohort is 4.5% default and 75.0% prepaid; the crisis cohort, 10.5% and 87.1%. Default is the first month at 90 or more days past due, following [4]; prepayment is zero-balance code 01. Time is loan age in months, so the origin is origination. These data reproduce the contrast that motivates the competing-risks framing: counting prepayment as an event leaves 20.6% of post-crisis loans censored, while censoring it leaves 95.5%. Dirick et al. [7] report 20–85% and 94.56–98.16% on ten European consumer-credit portfolios; ours sit at the lower edge of the first range and inside the second, on a different product, country and decade. Prepayment is not a nuisance here but the dominant outcome, outnumbering default seventeen to one.

B. Models and the common interface

Every model implements the same interface, taking covariates, times and event labels and returning a predicted cumulative incidence per cause on a fixed horizon grid, so no model can be given a metric the others cannot. We compare the Aalen–Johansen estimator as a null model that cannot discriminate but is calibrated by construction, a cause-specific Cox model, a Fine–Gray subdistribution model, a random survival forest and DeepHit. Two implementation details matter for interpretation. Cox and Fine–Gray were implemented directly, since no mainstream Python library provides Fine–Gray; both maximise a Breslow partial likelihood, with Fine–Gray retaining subjects who experience the competing event in the risk set under the censoring weights of [1]. Our forest is a cause-specific random survival forest, fitting one forest per cause, and is therefore a relative of rather than a replication of the composite-splitting competing-risks forest used by Frydman and Matuszyk [2].

C. Metrics

Discrimination is the inverse-probability-of-censoring-weighted time-dependent AUC at 12 to 60 months, with subjects who experience the competing event treated as controls because they can never experience the cause of interest. Calibration is measured in two ways: the integrated Brier score for the cumulative incidence, and a calibration index comparing mean predicted risk against the Aalen–Johansen estimate within deciles of predicted risk — a binned form of the integrated calibration index [17], whose competing-risks version smooths the curve rather than binning it [18]. Cost is measured fit and prediction time. Censoring weights come from a Kaplan–Meier estimate fitted on the training set only; because the time origin is origination and every vintage has at least eight years of follow-up, $G(60) = 0.9986$ in both cohorts, so those weights are near one and it is the competing event, not censoring, that shapes the results.

D. Splits, tuning budget and hypotheses

Models are trained on earlier origination years and tested on later ones, which is how a lender uses them, in contrast to the random splits that dominate the literature [4]. The post-crisis cohort trains on 2013–2015 and tests on 2016–2018; the crisis cohort trains on 2005–2006 and tests on 2007–2009. Because one such split yields a single number, we also evaluate by rolling origin: the training window expands one vintage at a time and the next vintage becomes the test set, giving four time-ordered folds per cohort. Hyperparameters were searched with a budget fixed in advance: twenty random configurations for DeepHit and eight for the forest, scored on an out-of-time validation vintage held inside the training years, so test vintages were never touched during tuning. Cox and Fine–Gray have no comparable hyperparameters and are reported untuned. Each configuration was scored on validation AUC and on validation integrated Brier score, and the winner under each criterion was refitted and evaluated separately.

For inference we fixed, before examining any test result, a two-sided null hypothesis that the models do not differ, a significance level of 0.05, and a Bonferroni correction across the three metric families, giving a threshold of 0.0167. A family is the three metrics of one pre-specified comparison in one cohort; thirty-six tests are computed in all, so the correction holds within a family, not across the table. Differences are assessed by a paired bootstrap rather than a parametric test, whose assumptions fail here: observations are paired, since the quantities compared come from the same test loans, and the sampling distributions of AUC and the calibration index are neither normal nor known in closed form. With 500 resamples it cannot resolve p below $1/500$; 0.002 is an upper bound. Because the bootstrap holds the fitted model fixed and understates total uncertainty, DeepHit was refitted with five seeds per cohort; its seed-to-seed

TABLE I

DISCRIMINATION (AUC AT 60 MONTHS) AND CALIBRATION (DECILE-BINNED ICI AT 60 MONTHS) FOR DEFAULT, ON THE HELD-OUT TEST VINTAGES OF EACH COHORT. THESE ARE SINGLE-SPLIT FIGURES, GIVEN FOR THE LEVELS; SECTION V-A REPORTS HOW FAR THEY MOVE ACROSS TIME-ORDERED FOLDS, AND THE DIFFERENCES BETWEEN MODELS HERE SHOULD BE READ AGAINST THAT SPREAD RATHER THAN ON THEIR OWN.

	Crisis (2007–09)		Post-crisis (2016–18)	
Model	AUC	ICI	AUC	ICI
Aalen–Johansen	0.5000	0.0077	–	–
Cox	0.8346	0.0169	0.7372	0.0278
Fine–Gray	0.8327	0.0159	–	–
RSF	0.8516	0.0286	0.7300	0.0287
RSF (tuned)	0.8538	0.0267	0.7275	0.0289
DeepHit α=0.2	0.8301	0.0281	0.7239	0.0242
DeepHit α=1	0.8405	0.0240	0.7246	0.0318
DeepHit (tuned)	0.8538	0.0209	0.7280	0.0318

standard deviation was 0.0014 and 0.0010 in AUC, and we treat differences below twice that as indistinguishable regardless of p.

V. RESULTS

A. No model leads reliably once the split is not a single draw

A single time-ordered split gives one number and no sense of how much that number moves. Expanding the training window one vintage at a time and testing on the next gives four folds per cohort instead (Table II).

Across the folds no model leads reliably. DeepHit's advantage over the cause-specific Cox model averages $+0.0073$ AUC on crisis defaults with a standard deviation of 0.0077 across folds, and the four fold-level values are $+0.0013$, $+0.0034$, $+0.0061$ and $+0.0185$: the largest is roughly three times the smallest. Against the forest the averages are smaller still. In all eight cause-by-cohort cells the ratio of the mean advantage to its spread across folds lies between -1.24 and $+0.95$, so none would survive a two-sided test at any conventional level. The same four models evaluated on one split per cohort had produced an apparently significant $+0.0192$ on crisis defaults; that figure is the 2009 fold, and it is the most extreme of the four.

B. The advantage comes from training alongside the test loans

If no model leads under time ordering, the question becomes what produces the lead that the literature reports. A random split changes two things at once: it lets a model train on loans originated alongside the test loans, and it changes which loans are tested. We separate them

TABLE II

ROLLING-ORIGIN EVALUATION. THE TRAINING WINDOW EXPANDS ONE VINTAGE AT A TIME AND THE NEXT VINTAGE IS THE TEST SET, GIVING FOUR TIME-ORDERED FOLDS PER COHORT. ENTRIES ARE DEEPHIT'S AUC ADVANTAGE AT 60 MONTHS OVER EACH BASELINE, AVERAGED OVER FOLDS, WITH THE STANDARD DEVIATION ACROSS FOLDS. NO ADVANTAGE REACHES 1.3 STANDARD DEVIATIONS OF ITS OWN SPREAD.

		vs Cox		vs RSF	
Cause	Cohort	mean	sd	mean	sd
def.	crisis	+0.0073	0.0077	+0.0023	0.0122
def.	post	-0.0043	0.0034	+0.0006	0.0012
prep.	crisis	+0.0050	0.0127	-0.0046	0.0295
prep.	post	+0.0083	0.0159	+0.0044	0.0092

by holding the test set fixed. A random half of the later vintages is the test set in both arms; one arm trains on the earlier vintages, the other on the remaining half of the later ones, subsampled to the same size. Only the training period differs (Table III).

Training on contemporaneous loans is worth between $+0.006$ and $+0.130$ AUC, and the gain is larger for DeepHit than for the Cox model in three of the four cause-by-cohort cells, by a factor of three on post-crisis default. The consequence is visible in the comparison that matters: against the tuned forest, DeepHit's advantage is positive in all four cells when training is contemporaneous and negative in three of four, negligible in the fourth, when it is not. On crisis prepayment the swing is from $+0.0091$ to -0.0696.

Both effects survive inference. Because the two arms predict the same loans, a paired bootstrap over those loans resamples them together; with $B = 500$ and the same Bonferroni threshold used elsewhere, every one of the sixteen contemporaneity bonuses is positive with an interval clear of zero, and DeepHit's is the largest of the four models in three of the four cells. The sharper quantity is the difference in differences — the change in DeepHit's lead over a baseline when training moves from earlier vintages to contemporaneous ones — which is significant in all eight comparisons. Against the forest it is positive in every cell, from $+0.0084$ to $+0.0787$, so the improvement in DeepHit's relative standing is not a sampling artefact. Against the Cox model it is positive in three cells and negative on crisis default, the one cell where the classical model gains more from contemporaneity than the deep one does.

The flexible model is not better at predicting forward. It is better at exploiting structure specific to the period it was trained on, and an evaluation protocol that supplies that structure will report an advantage that a lender scoring next year's originations does not receive.

TABLE III

THE SAME TEST LOANS, THE SAME TRAINING-SET SIZE, AND ONLY THE TRAINING PERIOD MOVED. THE TEST SET IS A RANDOM HALF OF THE LATER VINTAGES; ONE ARM TRAINS ON THE EARLIER VINTAGES, THE OTHER ON THE REMAINING HALF OF THE LATER ONES. ENTRIES ARE DEEPHIT'S AUC ADVANTAGE AT 60 MONTHS. AGAINST THE FOREST THE ADVANTAGE IS POSITIVE IN EVERY CELL WHEN TRAINING IS CONTEMPORANEOUS AND NEGATIVE OR NEGLIGIBLE IN EVERY CELL WHEN IT IS NOT. THE CHANGE IN EACH ADVANTAGE BETWEEN THE TWO ARMS IS SIGNIFICANT UNDER A PAIRED BOOTSTRAP OVER THE SHARED TEST LOANS ($B = 500$, BONFERRONI $0.05/3$).

		vs Cox		vs RSF	
Cause	Cohort	contemp.	earlier	contemp.	earlier
def.	crisis	+0.0019	+0.0064	+0.0048	-0.0060
def.	post	-0.0016	-0.0159	+0.0021	-0.0078
prep.	crisis	+0.0141	-0.0051	+0.0091	-0.0696
prep.	post	+0.0125	-0.0016	+0.0123	+0.0038

TABLE IV

POST-HOC RECALIBRATION, ICI AT 60 MONTHS. "TRAIN" ESTIMATES THE RESCALING ON THE TRAINING VINTAGES, WHICH IS ALL A LENDER HAS AT SCORING TIME; "ORACLE" ESTIMATES IT ON THE TEST VINTAGES AND IS UNATTAINABLE IN PRACTICE. COX AND DEEPHIT ARE SHOWN; THE FOREST AND FINE–GRAY BEHAVE THE SAME WAY. AUC IS IDENTICAL TO FOUR DECIMALS FOR RAW AND RECALIBRATED PREDICTIONS IN ALL SIXTEEN CELLS, CONFIRMING THE RESCALING IS MONOTONE.

Cohort	Cause	Model	Raw	Recal. train	Recal. oracle
crisis	def.	Cox	0.0169	0.0182	0.0061
crisis	def.	DeepHit	0.0209	0.0164	0.0053
crisis	prep.	Cox	0.2186	0.2204	0.1168
crisis	prep.	DeepHit	0.2361	0.2222	0.1063
post	def.	Cox	0.0278	0.0279	0.0025
post	def.	DeepHit	0.0318	0.0297	0.0082
post	prep.	Cox	0.1640	0.1693	0.0253
post	prep.	DeepHit	0.1850	0.1685	0.0439

C. Calibration is worse in every condition, and rescaling does not repair it

Where discrimination gives no stable ordering, calibration does. DeepHit's calibration index is 0.0040 worse than the Cox model's in both cohorts. The two carry that with very different force — about eighteen times the models' combined training variability post-crisis against 1.7 times on crisis data — so the sign replicates and the size does not. Every curve in Fig. 1 lies above the diagonal: trained on 2005–06 and tested on 2007–09, all three models under-predict, and the deciles separate by a few percentage points for default against twenty or more for prepayment.

Following Alberge et al. [13], who show that recalibration can improve competing-risks probabilities while preserving discrimination, we estimated a marginal per-horizon rescaling two ways (Table IV). Fitted on the training vintages, which is all a lender has when scoring a new loan, it changes almost nothing and in ten of sixteen cells makes calibration worse. Fitted on the test vintages it cuts the index by a median factor of 4.2, reaching 14.7 for Fine–Gray on post-crisis default. The penalty is therefore not a defective calibration map, or the oracle correction would not work; it is the vintage shift, since a rescaling estimated on 2005–06 cannot know that 2007–09 would turn out worse. Recalibration as demonstrated in the literature is an in-distribution procedure, and extending it across vintages is an open problem rather than a step we omitted.

D. Cost, and two checks that changed nothing

The models span three orders of magnitude in training cost. On 150,000 loans Fine–Gray fits in 0.03 seconds and the cause-specific Cox model in 0.42, against 36 seconds for the forest and 59 for DeepHit, and the DeepHit search added 14.4 minutes across the two cohorts against 4.1 for the forest, while the classical models have no search to run. Since the deep model leads nothing reliably, that factor of roughly two thousand buys no measured improvement. We report these figures because the benchmarking literature generally omits them [6].

Two pre-registered checks returned nothing, and both are worth recording. First, we expected DeepHit's ranking-loss weight to matter: pooling our forty tuning configurations by α suggests validation AUC rising from 0.658 at $\alpha = 0.1$ to 0.776 at $\alpha = 1.0$. Holding architecture, dropout, learning rate, batch size and bin count fixed and varying only α over five seeds reverses that picture — the change is $+0.0075$, $+0.0005$, -0.0673 and $+0.0386$ across the four cells against seed standard deviations of 0.0033, 0.0018, 0.0103 and 0.0062 — so the effect is inconsistent in sign and no setting can be recommended. A marginal taken over a random search is not an ablation. Second, we expected that selecting a configuration on validation discrimination would cost calibration, and it did not: three of four model-by-cohort comparisons differ by less than twice DeepHit's seed-to-seed variation, and the one robust effect is a $+0.0101$ AUC gain for the configuration chosen on AUC, which is selection working as intended.

VI. DISCUSSION, LIMITATIONS AND CONCLUSION

Taken together these results qualify, rather than contradict, the finding that deep competing-risks models perform well on mortgage data [4]. They do perform well under the protocol that literature uses. Under rolling origin the advantage does not survive, and the matched

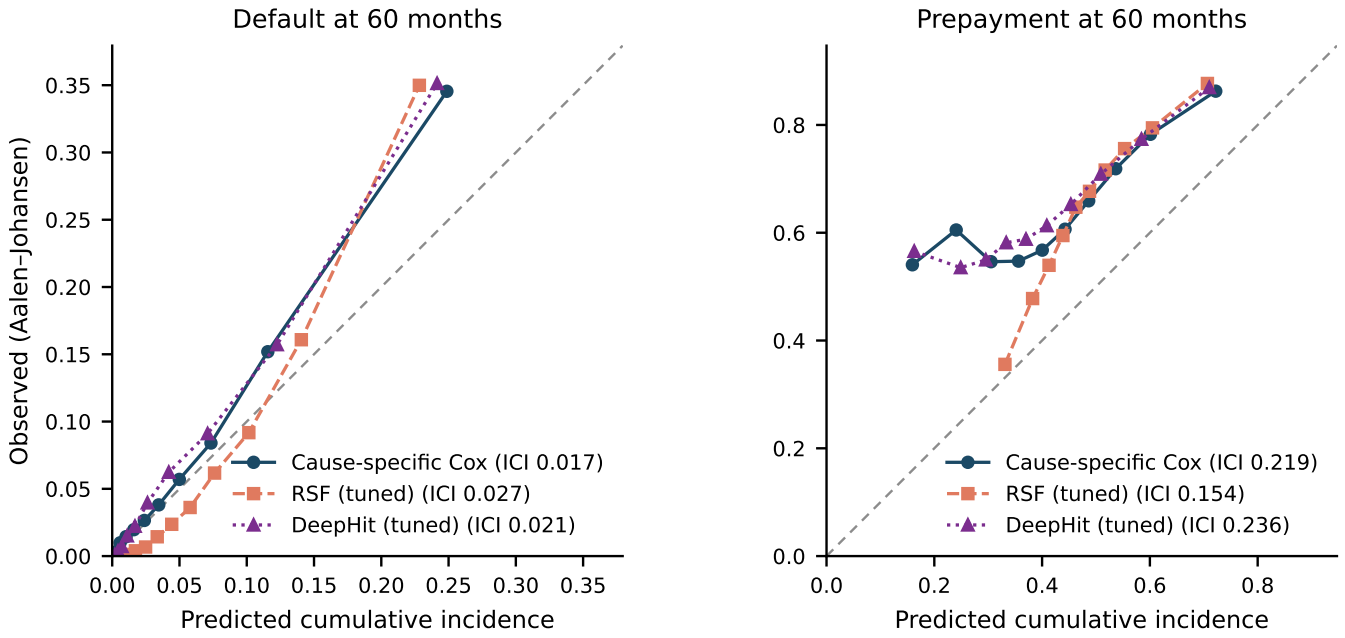


Fig. 1. Calibration at 60 months on the crisis cohort (train 2005–06, test 2007–09). Each point is a decile of predicted risk; the vertical axis is the Aalen–Johansen estimate within that decile, so competing events and censoring are both accounted for. A model on the diagonal is perfectly calibrated; every curve lies above it, so all three under-predict out of time. The deciles separate by a few percentage points for default and by twenty or more for prepayment, the competing event the credit literature most often censors.

comparison says why: what produces it is the contemporaneity of the training data rather than the model class. An untuned deep baseline is not evidence, and neither is a tuned one evaluated on loans originated beside its training set. For a lender who needs a probability rather than an ordering, and who scores loans that do not yet exist, three orders of magnitude of extra compute are difficult to justify.

Against the quality criteria for quantitative work, reliability is measured rather than asserted: every comparison carries a variability estimate from seeds, from bootstrap resamples of the test set, and from refitting on resampled training vintages. Replicability is the public pipeline. Internal validity rests on varying one factor at a time, which the α ablation and the protocol comparison each enforce. Construct validity is the paper's subject rather than an assumption: we ask whether discrimination measures what it is used to decide. Generalisation is weakest, and bounded by the first delimitation above.

Within those bounds, several limitations remain. Our recalibration is a marginal per-horizon rescaling rather than the per-cause procedures of Alberge et al. [13]. Training variability is estimated from five bootstrap resamples, which conflates sampling with optimisation noise. The forest is cause-specific rather than Ishwaran's competing-risks formulation. And our covariates are observed at origination, so we cannot test directly whether the period-specific structure the deep model exploits is the macroeconomic variation that Bellotti and Crook [10] model explicitly.

Under a protocol that measures ranking, calibration and cost together on time-ordered splits, deep competing-risks models offer no reliable advantage over classical or tree-based baselines on mortgage data, and are worse calibrated in every condition we tested. What generalises past this benchmark is the mechanism rather than the verdict: the advantage these models are reported to have is substantially a property of being evaluated on loans contemporaneous with their training data, and the same caution applies to any flexible model assessed that way.

Author Contributions

The three reviews were written independently: A.S. on competing risks in credit scoring, A.E.N. on calibration, F.B. on evaluation protocol and cost. A.S. built the pipeline, models and metrics and ran the experiments; A.E.N. specified the calibration measures and recalibration study; F.B. the split protocol, tuning budget and cost measurement. The gap, hypotheses and final analysis were agreed jointly; code at github.com/ar-sadraei/rtds-project.

REFERENCES

[1] J. P. Fine and R. J. Gray, "A proportional hazards model for the subdistribution of a competing risk," *Journal of the American Statistical Association*, vol. 94, no. 446, pp. 496–509, 1999.

[2] H. Frydman and A. Matuszyk, "Random survival forest for competing credit risks," *J. Oper. Res. Soc.*, vol. 73, no. 1, pp. 15–25, 2022. [Online]. Available: https://doi.org/10.1080/01605682.2020.1759385

[3] C. Lee, W. R. Zame, J. Yoon, and M. van der Schaar, "Deephit: A deep learning approach to survival analysis with competing risks," in *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, (AAAI-18), the 30th innovative Applications of Artificial Intelligence (IAAI-18), and the 8th AAAI Symposium on Educational Advances in Artificial Intelligence (EAAI-18), New Orleans, Louisiana, USA, February 2-7, 2018*, S. A. McIlraith and K. Q. Weinberger, Eds. AAAI Press, 2018, pp. 2314–2321. [Online]. Available: https://doi.org/10.1609/aaai.v32i1.11842

[4] G. Blumenstock, S. Lessmann, and H. Seow, "Deep learning for survival and competing risk modelling," *J. Oper. Res. Soc.*, vol. 73, no. 1, pp. 26–38, 2022. [Online]. Available: https://doi.org/10.1080/01605682.2020.1838960

[5] L. Shang, J. Zhao, G. Li, and X. Zhang, "Survival analysis in credit risk management: a review study," *The Journal of Credit Risk*, vol. 20, no. 4, 2024.

[6] S. Wiegrebe, P. Kopper, R. Sonabend, B. Bischl, and A. Bender, "Deep learning for survival analysis: a review," *Artificial Intelligence Review*, vol. 57, no. 3, p. 65, 2024.

[7] L. Dirick, G. Claeskens, and B. Baesens, "Time to default in credit scoring using survival analysis: a benchmark study," *J. Oper. Res. Soc.*, vol. 68, no. 6, pp. 652–665, 2017. [Online]. Available: https://doi.org/10.1057/s41274-016-0128-9

[8] M. Thackham and J. Ma, "On maximum likelihood estimation of competing risks using the cause-specific semi-parametric cox model with time-varying covariates - an application to credit risk," *J. Oper. Res. Soc.*, vol. 73, no. 1, pp. 5–14, 2022. [Online]. Available: https://doi.org/10.1080/01605682.2020.1800418

[9] C. A. P. B. Saavedra, J. B. F. Gomes, E. M. de Castro Gomes, and H. Kimura, "Probability of default for lifetime credit loss for IFRS 9 using machine learning competing risks survival analysis models," *Expert Syst. Appl.*, vol. 249, p. 123607, 2024. [Online]. Available: https://doi.org/10.1016/j.eswa.2024.123607

[10] T. Bellotti and J. Crook, "Credit scoring with macroeconomic variables using survival analysis," *J. Oper. Res. Soc.*, vol. 60, no. 12, pp. 1699–1707, 2009. [Online]. Available: https://doi.org/10.1057/jors.2008.130

[11] H. Haider, B. Hoehn, S. Davis, and R. Greiner, "Effective ways to build and evaluate individual survival distributions," *J. Mach. Learn. Res.*, vol. 21, pp. 85:1–85:63, 2020. [Online]. Available: https://jmlr.org/papers/v21/18-772.html

[12] R. Sonabend, A. Bender, and S. J. Vollmer, "Avoiding c-hacking when evaluating survival distribution predictions with discrimination measures," *Bioinform.*, vol. 38, no. 17, pp. 4178–4184, 2022. [Online]. Available: https://doi.org/10.1093/bioinformatics/btac451

[13] J. Alberge, T. Haugomat, G. Varoquaux, and J. Abécassis, "On the calibration of survival models with competing risks," in *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, ser. Proceedings of Machine Learning Research, E. Khan, Y. Li, A. Solin, and A. Ramdas, Eds., vol. 300. PMLR, 2026, pp. 1837–1845. [Online]. Available: https://proceedings.mlr.press/v300/alberge26a.html

[14] M. Bank and B. Eder, "A review on the probability of default for IFRS 9," SSRN, Tech. Rep., 2021.

[15] A. Botha and T. Verster, "Approaches for modelling the term-structure of default risk under IFRS 9: a tutorial using discrete-time survival analysis," *International Journal of Data Science and Analytics*, vol. 22, no. 1, 2026.

[16] E. Wycinka, "Competing risk models of default in the presence of early repayments," *Econometrics*, vol. 23, no. 2, pp. 99–120, 2019.

[17] P. C. Austin and E. W. Steyerberg, "The integrated calibration index (ICI) and related metrics for quantifying the calibration of logistic regression models," *Statistics in Medicine*, vol. 38, no. 21, pp. 4051–4065, 2019.

[18] P. C. Austin, H. Putter, D. Giardiello, and D. van Klaveren, "Graphical calibration curves and the integrated calibration index (ICI) for competing risk models," *Diagnostic and Prognostic Research*, vol. 6, no. 1, p. 2, 2022.

[19] H. Kvamme, Ø. Borgan, and I. Scheel, "Time-to-event prediction with neural networks and cox regression," *J. Mach. Learn. Res.*, vol. 20, pp. 129:1–129:30, 2019. [Online]. Available: https://jmlr.org/papers/v20/18-424.html